Finding Building Blocks for a Target Molecule

A visual explanation of the single-step precursor recommendation workflow

Objective

Given P_target and a bank of compounds, find small sets of building blocks whose conserved molecular pieces can be combined to account for the complete target.

Workflow

1. Search the bank

Compare every building block with P_target.

→

2. Find useful pieces

Retain coherent fragments that occur in the target.

→

3. Assemble coverage

Combine complementary pieces until P_target is covered.

→

4. Show alternatives

Return the best candidate from each distinct pattern.

The next slides show the result directly as R1 + R2 to P_target mappings.

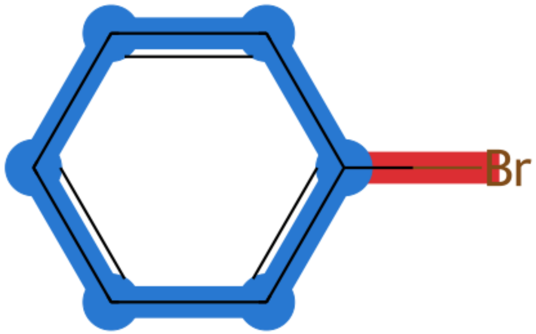
Matching colors come from computed atom correspondences, not manual annotation.

The output is a structural recommendation; chemical feasibility still requires review.

Example 1: two building blocks cover P_target

Complete 2D molecules are shown; identical colors are assigned by the computed atom mapping

R1: bromobenzene



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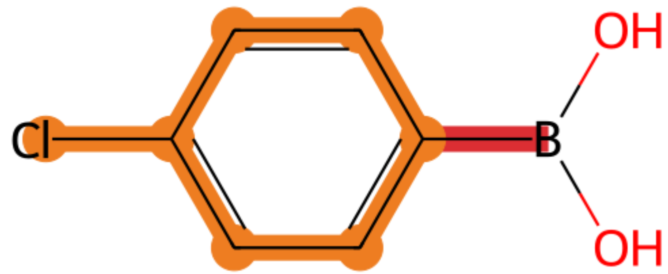
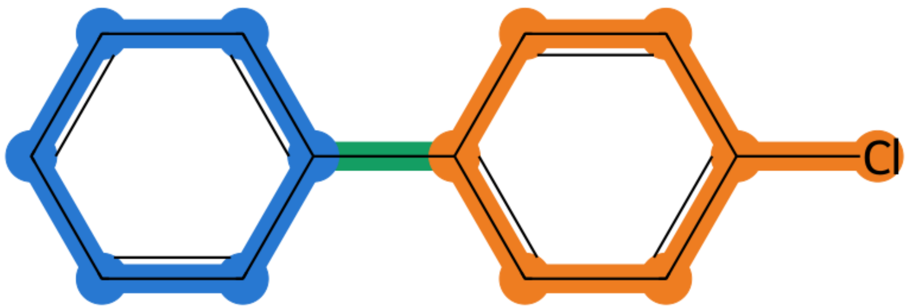
R2: 4-chlorophenylboronic acid



P_target: 4-chlorobiphenyl



What the mapping shows

- blue atoms come from R1
- orange atoms come from R2
- red source groups are not retained
- green is the new bond joining the pieces

Example 2: alternative building-block patterns

One best representative is shown for each distinct structural division of phenylacetic acid

Pattern A

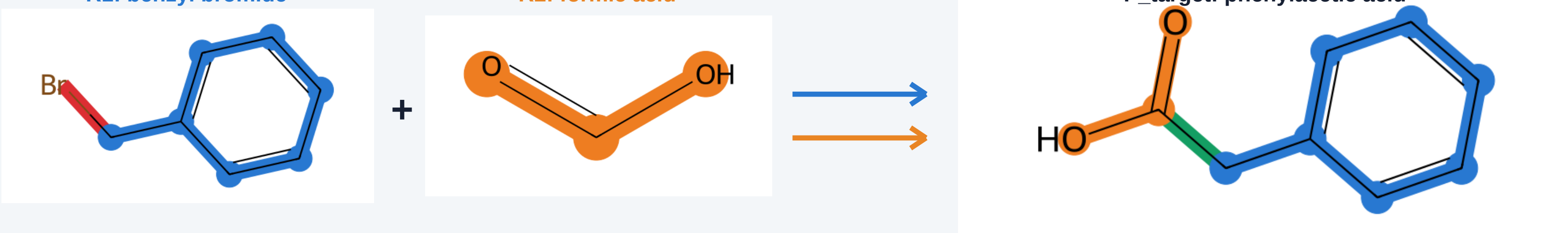
carbonyl framework retained

R1: phenylacetyl chloride | R2: water | P_target: phenylacetic acid

Pattern B

benzyl and carboxyl modules

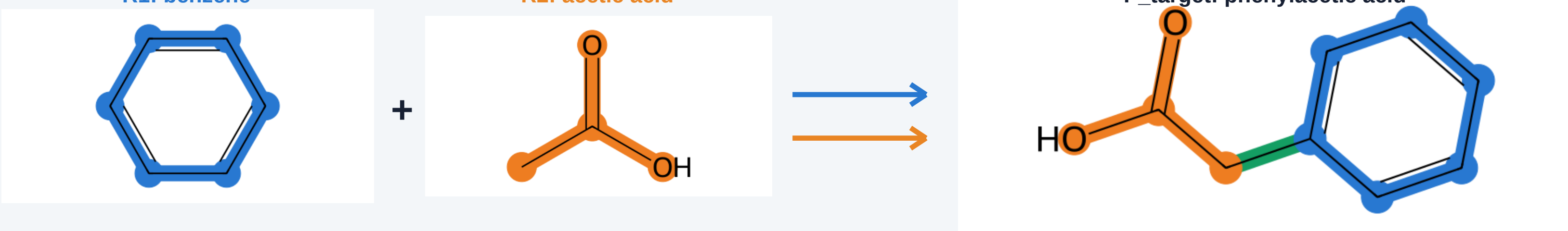
R1: benzyl bromide | R2: formic acid | P_target: phenylacetic acid



Pattern C

aryl and acetic-acid modules

R1: benzene | R2: acetic acid | P_target: phenylacetic acid



- Every row completely covers the same P_target, but divides it between R1 and R2 differently.
- The workflow returns one representative per pattern; chemical feasibility is reviewed separately.

Example 3: one building block contributes multiple pieces

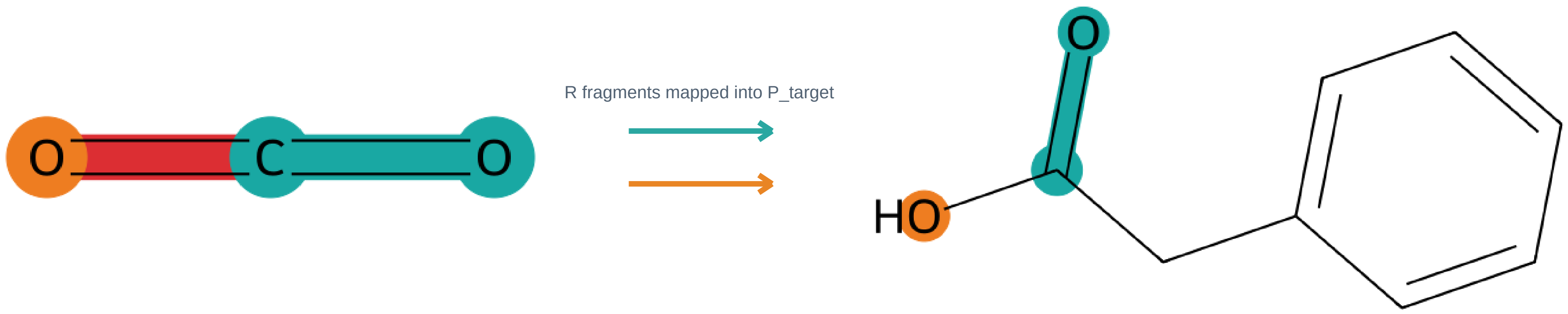
CO2 is recognized within the carboxyl group of phenylacetic acid

R: carbon dioxide

The carbonyl piece is recognized first. The second oxygen is then retained as another useful piece of the same source.

P_target: phenylacetic acid

Only the colored carboxyl atoms are owned by this candidate. The aromatic side of P remains available to other sources.



- initial retained fragment
- additional target-owned fragment

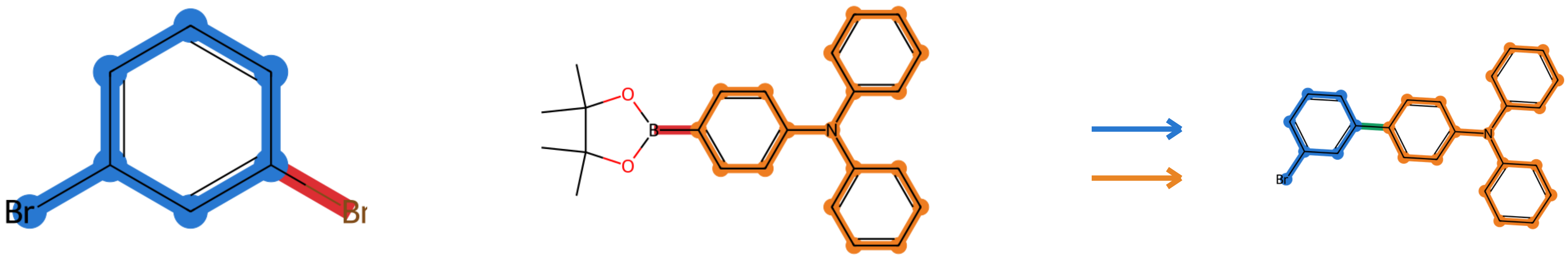
What the colors show

- The cyan carbonyl is the main connected match.
- The orange oxygen is recognized as a second useful piece from the same R.
- Together, the colored atoms account for the CO2 contribution; the uncolored target remains available to other building blocks.

Example 4: the same workflow on a larger target

R1 and R2 are mapped into a complete P_target using the same color convention

R1: 1,3-dibromobenzene **R2: triarylamine BPin** **P_target: assembled product**



● retained smaller aryl module ● retained triarylamine module ● new inter-module bond

What this example demonstrates

- The large triarylamine system remains one coherent orange contribution from R2.
- The smaller blue aryl module comes from R1.
- Unused bromine and the BPin unit are uncolored because they are not retained in P_target.
- Together, the blue and orange regions cover the complete target; green marks their new bond.